Demographics

Overview

Here is my initial exploration of participant demographics.

Here are my research questions:
- How many respondents took the survey, and from how many institutions?
- Which regions are those institutions located in?
- What roles do our respondents represent?
- What are our respondents' professional backgrounds?

Import packages, utilities, and config variables

```
library(readxl)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(docstring)
```

```
Attaching package: 'docstring'

The following object is masked from 'package:utils':

    ?
```

```
library(readr)
```

```
Warning: package 'readr' was built under R version 4.4.3
```

```
source("utils.R")

# Read config.yml
cfg <- config::get()
```

Load data

```
mc_complex <- read_tsv(file.path(cfg$data_path, "mc_complex.tsv"), show_col_types = FALSE, sk
mc_other <- read_tsv(file.path(cfg$data_path, "mc_other.tsv"), show_col_types = FALSE, skip_e
prof_bg <- read_tsv(file.path(cfg$data_path, "prof_bg.tsv"), show_col_types = FALSE, skip_emp
sata_other <- read_tsv(file.path(cfg$data_path, "sata_other.tsv"), show_col_types = FALSE, sk
qual <- read_tsv(file.path(cfg$data_path, "qual.tsv"), show_col_types = FALSE, skip_empty_row
qual_pii <- read_tsv(file.path(cfg$data_path, "qual_pii.tsv"), show_col_types = FALSE, skip_e
```

How many respondents took the survey?

```
nrow(mc_complex)
```

```
[1] 33
```

```
nrow(prof_bg)
```

```
[1] 33
```

The numbers above^ should match, and this represents the number of respondents.

How many institutions?

I'm commenting out the relevant code for privacy reasons, but I first manually inspected the responses.

```
# Keep commented for privacy!!
# qual_pii$institution
```

Looking at the responses, I see several "categories" of institution: Network, networked university campus, independent university campus, and university-affiliated research center. "Networked" refers specifically to an OSPO network, not any other kind of network. Grouping the institutions by category, rather than listing them, in the final report will help protect respondent anonymity.

I have manually cleaned up the responses and put them in a manually-curated excel file. The first sheet has two columns: the participant's response to the organization question and my standardized name for the org. The second sheet lists the category and region associated with that org.

```
# Do not preview in notebook, for privacy!!
org_responses <- read_excel(file.path(cfg$curated_path, "orgs.xlsx"), sheet="responses")
org_info <- read_excel(file.path(cfg$curated_path, "orgs.xlsx"), sheet="org_info")

# Make sure the manually curated spreadsheet matches the survey data tsv
identical(qual_pii$institution, org_responses$response)
```

```
[1] TRUE
```

```
# This is the number of unique institutions
length(unique(org_responses$organization))
```

```
[1] 27
```

```r
merged <- left_join(org_responses, org_info, by = "organization")

# Keeps the FIRST occurrence of each unique value in 'organization'
# Use this df to look at the breakdown of institutions, where each
# institution is counted only once.
merged_unique <- merged |>
  distinct(organization, .keep_all = TRUE)

# This is the breakdown of institutions by category
table(merged_unique$category)
```

```
      Independent Research Center     Independent University Campus
                                2                                14
                          Network       Networked University Campus
                                1                                 6
                            Other University-hosted Research Center
                                1                                 3
```

Which regions are those institutions located in?

```r
table(merged_unique$region)
```

```
Europe     US
     8     19
```

^The table above shows represented regions.

Examining what people said for “other”

```r
# Keep commented for privacy!!
# mc_other$region_other
```

What roles do our respondents represent?

```
role_label <- "role_mc"
table(mc_complex[[role_label]])
```

```
   I am a staff member of an OSPO or equivalent function
                                                       8
I carry out OSPO-like activities as part of another role
                                                       6
        I lead or co-lead an OSPO or equivalent function
                                                      18
                                                   Other
                                                       1
```

Examining what people said for "other"

```
# Keep commented for privacy!!
# mc_other$role_other
```

What about job titles? Need to code these. Writing to a file for this purpose.

```
write_df_to_file(qual$job, "job_titles.txt", path_key = "coding_path")
```

What are our respondents' professional backgrounds?

```
# Returns a list containing the frequency table for each column
table_list <- lapply(prof_bg, table)

# Examined the output by eye to make sure there's no weirdness
# Keep commented for privacy
# table_list

# Extract the "Yes" count for each professional background
# This was a SATA question so total != # of respondents
yes_counts <- sapply(table_list, function(x) x["Yes"])
yes_counts <- sort(yes_counts, decreasing = TRUE)
print(yes_counts)
```

```
     rse.Yes   faculty.Yes     admin.Yes       tto.Yes        it.Yes
          10             9             8             7             7
industry.Yes     legal.Yes community.Yes   library.Yes
           5             4             4             2
```

Examining the "other" responses. Mostly it looks like people just elaborating on the responses above, though one person said they were previously a volunteer to OS projects, which is interesting.

```
# Keep commented for privacy!!
# sata_other$prof_bg_other
```